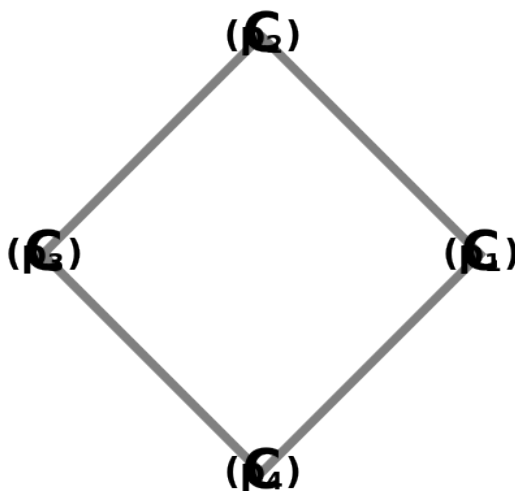


Cyclopentadiene



Point Group: **D_{4h}**

p orbitals

Symmetry behavior of the p-orbitals:

$$\Gamma_{red} = 4E ; -2C'2 ; -4\sigma h ; 2\sigma v$$

$$\Gamma_{irred} = 1Eg \oplus 1A2u \oplus 1B2u$$

- SALC of Eg: $\frac{\sqrt{2}(p_1 - p_3)}{2}$
- SALC of A2u: $\frac{p_1}{2} + \frac{p_2}{2} + \frac{p_3}{2} + \frac{p_4}{2}$
- SALC of B2u: $\frac{p_1}{2} - \frac{p_2}{2} + \frac{p_3}{2} - \frac{p_4}{2}$
- SALC of Eg: $\frac{\sqrt{2}(p_2 - p_4)}{2}$

Put together these SALCs form the following Hamilton matrices:

$$H_{Eg} = \begin{pmatrix} \tilde{\alpha} & 0 \\ 0 & \tilde{\alpha} \end{pmatrix}$$

$$H_{A2u} = (\tilde{\alpha} + 2\tilde{\beta})$$

$$H_{B2u} = (\tilde{\alpha} - 2\tilde{\beta})$$

Solving Hückels secular equations, that follow from these H, leads to:

- orbital of A2u with energy = $\tilde{\alpha} + 2\tilde{\beta}$

(

$$\begin{aligned} (1) * (1. \text{ SALC in A2u}) &= (1) * \left(\frac{p_1}{2} + \frac{p_2}{2} + \frac{p_3}{2} + \frac{p_4}{2} \right) \\ &= \frac{p_1}{2} + \frac{p_2}{2} + \frac{p_3}{2} + \frac{p_4}{2} \end{aligned}$$

)

- orbital of Eg with energy = $\tilde{\alpha}$

(

$$\begin{aligned} \begin{pmatrix} 1 & 0 \end{pmatrix} * \begin{pmatrix} 1. \text{ SALC in Eg} \\ 2. \text{ SALC in Eg} \end{pmatrix} &= \begin{pmatrix} 1 & 0 \end{pmatrix} * \begin{pmatrix} \frac{\sqrt{2}p_1}{2} - \frac{\sqrt{2}p_3}{2} \\ \frac{\sqrt{2}p_2}{2} - \frac{\sqrt{2}p_4}{2} \end{pmatrix} \\ &= \frac{\sqrt{2}(p_1 - p_3)}{2} \end{aligned}$$

)

- orbital of Eg with energy = $\tilde{\alpha}$

(

$$\begin{aligned} \begin{pmatrix} 0 & 1 \end{pmatrix} * \begin{pmatrix} 1. \text{ SALC in Eg} \\ 2. \text{ SALC in Eg} \end{pmatrix} &= \begin{pmatrix} 0 & 1 \end{pmatrix} * \begin{pmatrix} \frac{\sqrt{2}p_1}{2} - \frac{\sqrt{2}p_3}{2} \\ \frac{\sqrt{2}p_2}{2} - \frac{\sqrt{2}p_4}{2} \end{pmatrix} \\ &= \frac{\sqrt{2}(p_2 - p_4)}{2} \end{aligned}$$

)

- orbital of B2u with energy = $\tilde{\alpha} - 2\tilde{\beta}$

(

$$\begin{aligned} (1) * (1. \text{ SALC in B2u}) &= (1) * \left(\frac{p_1}{2} - \frac{p_2}{2} + \frac{p_3}{2} - \frac{p_4}{2} \right) \\ &= \frac{p_1}{2} - \frac{p_2}{2} + \frac{p_3}{2} - \frac{p_4}{2} \end{aligned}$$

)

s orbitals

Symmetry behavior of the s-orbitals:

$$\Gamma_{red} = 4E ; 2C'2 ; 4\sigma_h ; 2\sigma_v$$

$$\Gamma_{irred} = 1A1g \oplus 1B1g \oplus 1Eu$$

- SALC of A1g: $\frac{s_1}{2} + \frac{s_2}{2} + \frac{s_3}{2} + \frac{s_4}{2}$
- SALC of B1g: $\frac{s_1}{2} - \frac{s_2}{2} + \frac{s_3}{2} - \frac{s_4}{2}$
- SALC of Eu: $\frac{\sqrt{2}(s_1 - s_3)}{2}$
- SALC of Eu: $\frac{\sqrt{2}(s_2 - s_4)}{2}$

Put together these SALCs form the following Hamilton matrices:

$$H_{A1g} = (\tilde{\alpha}_s + 2\tilde{\beta}_s)$$

$$H_{B1g} = (\tilde{\alpha}_s - 2\tilde{\beta}_s)$$

$$H_{Eu} = \begin{pmatrix} \tilde{\alpha}_s & 0 \\ 0 & \tilde{\alpha}_s \end{pmatrix}$$

Solving Hückels secular equations, that follow from these H, leads to:

- orbital of A1g with energy = $\tilde{\alpha}_s + 2\tilde{\beta}_s$

(

$$\begin{aligned} (1) * (1. \text{ SALC in A1g}) &= (1) * \left(\frac{s_1}{2} + \frac{s_2}{2} + \frac{s_3}{2} + \frac{s_4}{2} \right) \\ &= \frac{s_1}{2} + \frac{s_2}{2} + \frac{s_3}{2} + \frac{s_4}{2} \end{aligned}$$

)

- orbital of Eu with energy = $\tilde{\alpha}_s$

(

$$\begin{aligned} \begin{pmatrix} 1 & 0 \end{pmatrix} * \begin{pmatrix} 1. \text{ SALC in Eu} \\ 2. \text{ SALC in Eu} \end{pmatrix} &= \begin{pmatrix} 1 & 0 \end{pmatrix} * \begin{pmatrix} \frac{\sqrt{2}s_1}{2} - \frac{\sqrt{2}s_3}{2} \\ \frac{\sqrt{2}s_2}{2} - \frac{\sqrt{2}s_4}{2} \end{pmatrix} \\ &= \frac{\sqrt{2}(s_1 - s_3)}{2} \end{aligned}$$

)

- orbital of Eu with energy = $\tilde{\alpha}_s$

(

$$\begin{aligned} \begin{pmatrix} 0 & 1 \end{pmatrix} * \begin{pmatrix} 1. \text{ SALC in Eu} \\ 2. \text{ SALC in Eu} \end{pmatrix} &= \begin{pmatrix} 0 & 1 \end{pmatrix} * \begin{pmatrix} \frac{\sqrt{2}s_1}{2} - \frac{\sqrt{2}s_3}{2} \\ \frac{\sqrt{2}s_2}{2} - \frac{\sqrt{2}s_4}{2} \end{pmatrix} \\ &= \frac{\sqrt{2}(s_2 - s_4)}{2} \end{aligned}$$

)

- orbital of B1g with energy = $\tilde{\alpha}_s - 2\tilde{\beta}_s$

(

$$\begin{aligned} (1) * (1. \text{ SALC in B1g}) &= (1) * \left(\frac{s_1}{2} - \frac{s_2}{2} + \frac{s_3}{2} - \frac{s_4}{2} \right) \\ &= \frac{s_1}{2} - \frac{s_2}{2} + \frac{s_3}{2} - \frac{s_4}{2} \end{aligned}$$

)

Transitions from p orbitals into s orbitals

State with Occupation (2, 2, 0, 0) (0 unpaired electrons, mult 1.0)

Transition from ψ_2 into ψ_{s_2} :

- $-(2, 2, 0, 0), (0, 0, 0, 0) \rightarrow (2, 1, 0, 0), (0, 1, 0, 0)$
- ψ_2 : linear combination = $\sqrt{2}(\mathbf{p}_1 - \mathbf{p}_3)$, energy = $\tilde{\alpha}$
- η_2 : linear combination = $\sqrt{2}(\mathbf{s}_1 - \mathbf{s}_3)$, energy = $\tilde{\alpha}_s$
- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s$
- Energy of State before Excitation: $4\tilde{\alpha} + 4\tilde{\beta}$
- Energy of State after Excitation: $3\tilde{\alpha} + \tilde{\alpha}_s + 4\tilde{\beta}$
- Energy between Average Energies of ψ -/ η -orbitals: $-\tilde{\alpha} + \tilde{\alpha}_s$
- dipole transition moment: $\tilde{\delta}$

State with Occupation (2, 1, 1, 0) (2 unpaired electrons, mult 3.0)

Transition from ψ_2 into ψ_{s_2} :

- $-(2, 1, 1, 0), (0, 0, 0, 0) \rightarrow (2, 0, 1, 0), (0, 1, 0, 0)$
- ψ_2 : linear combination = $\sqrt{2}(\mathbf{p}_1 - \mathbf{p}_3)$, energy = $\tilde{\alpha}$
- η_2 : linear combination = $\sqrt{2}(\mathbf{s}_1 - \mathbf{s}_3)$, energy = $\tilde{\alpha}_s$
- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s$
- Energy of State before Excitation: $4\tilde{\alpha} + 4\tilde{\beta}$
- Energy of State after Excitation: $3\tilde{\alpha} + \tilde{\alpha}_s + 4\tilde{\beta}$
- Energy between Average Energies of ψ -/ η -orbitals: $-\tilde{\alpha} + \tilde{\alpha}_s$
- dipole transition moment: $\tilde{\delta}$

Transition from ψ_3 into ψ_{s_3} :

- $-(2, 1, 1, 0), (0, 0, 0, 0) \rightarrow (2, 1, 0, 0), (0, 0, 1, 0)$
- ψ_3 : linear combination = 0, energy = $\tilde{\alpha}$
- η_3 : linear combination = 0, energy = $\tilde{\alpha}_s$
- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s$
- Energy of State before Excitation: $4\tilde{\alpha} + 4\tilde{\beta}$
- Energy of State after Excitation: $3\tilde{\alpha} + \tilde{\alpha}_s + 4\tilde{\beta}$
- Energy between Average Energies of ψ -/ η -orbitals: $-\tilde{\alpha} + \tilde{\alpha}_s$
- dipole transition moment: $\tilde{\delta}$

- AND

Dispersion energies following from the transitions

1. State: $A = (2, 1, 1, 0) \leftrightarrow B = (2, 1, 1, 0)$

$$E_{dispersion} = -\frac{8\tilde{\delta}^4}{r^6 (\tilde{\alpha} - \tilde{\alpha}_s)}$$

Ground-State:

$A = [2, 2, 0, 0] \leftrightarrow B = [2, 2, 0, 0]$

$$E_{dispersion} = -\frac{8\tilde{\delta}^4}{r^6 (\tilde{\alpha} - \tilde{\alpha}_s)}$$